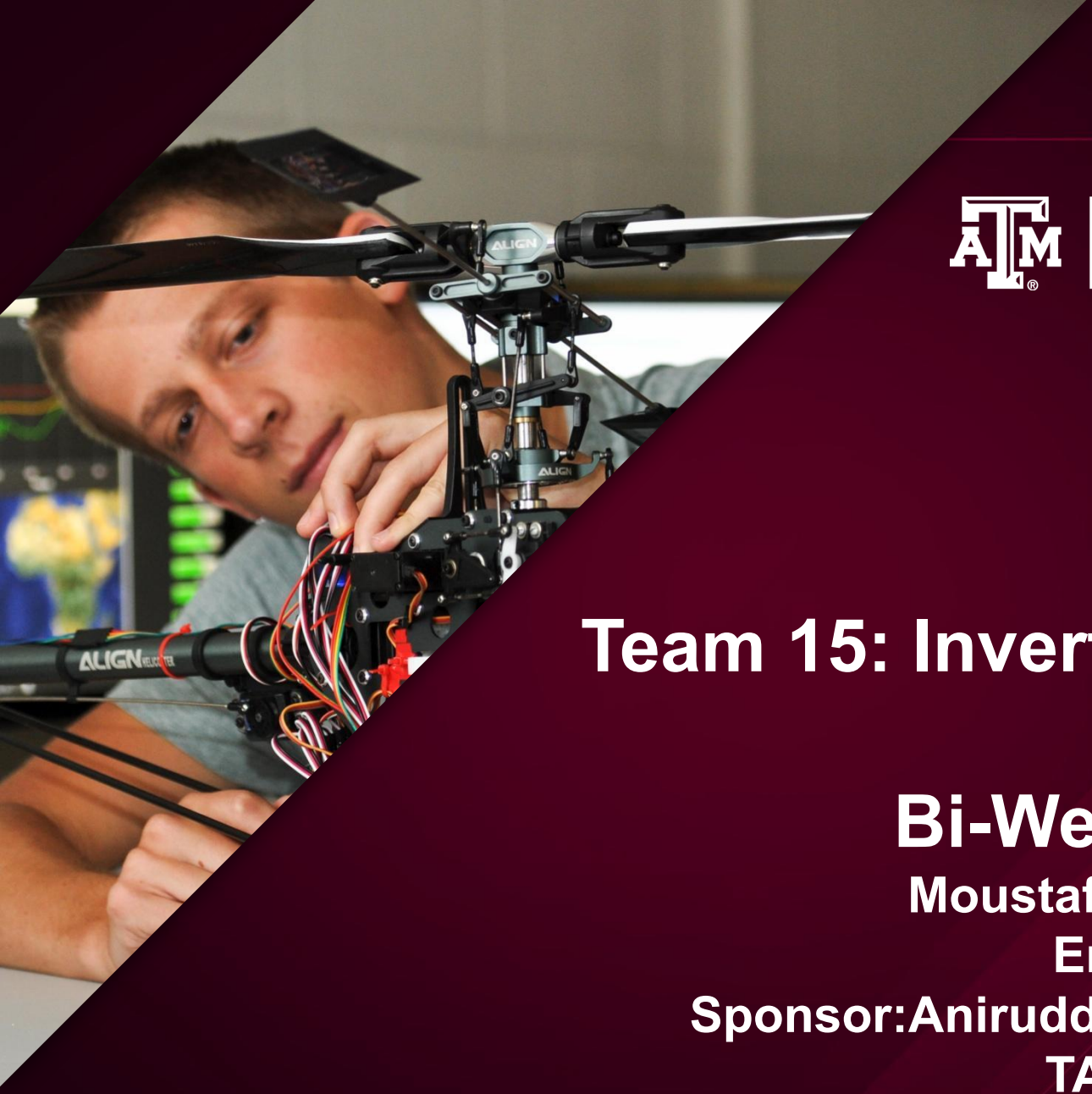

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Team 15: Inverted Pendulum Robot 1

Bi-Weekly Update 4

Moustafa Khalil, Kaitlynn Ly
Eric Chang, Gage Fox
Sponsor:Aniruddha Datta, Dr. Lusher
TA: Sabyasachi Gupta

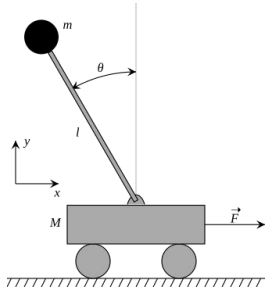
Project Summary

Problem statement:

- Autonomous warehouse robots generally experience instability when handling unbalanced loads or making sudden maneuvers. These movements can cause tilting, making balance a key challenge.

Our Solution Will:

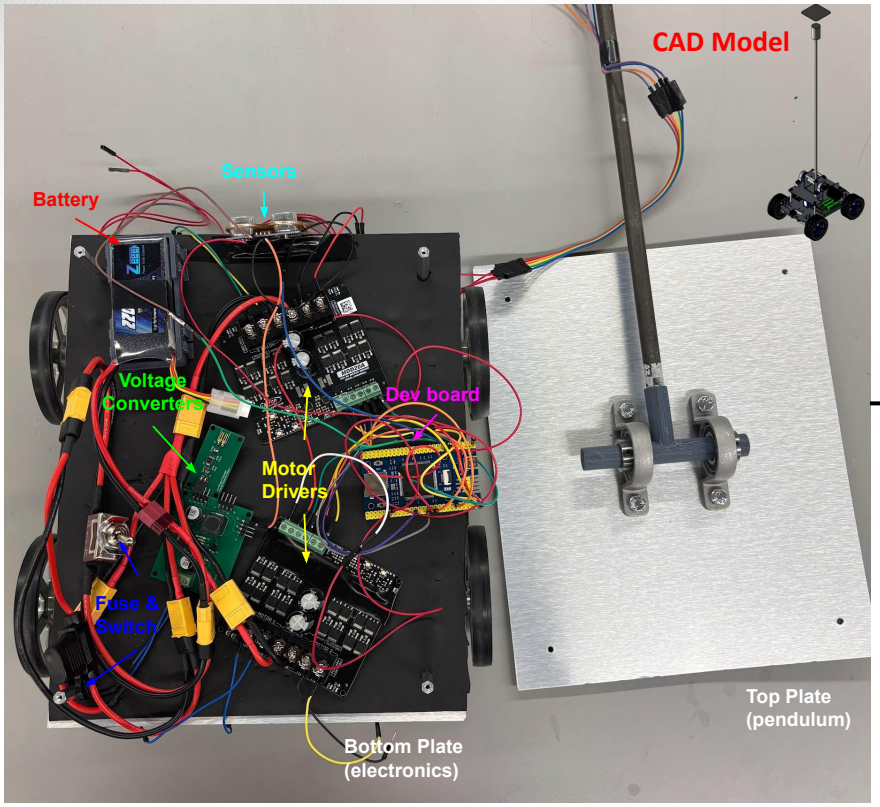
- Deliver an autonomous, self-navigating robot that maintains stability of an unstable payload while moving through constrained spaces.
- Integrate mechanical, sensing, power, and control subsystems into a cohesive, microcontroller-based platform designed for reliability and safety.



Side View

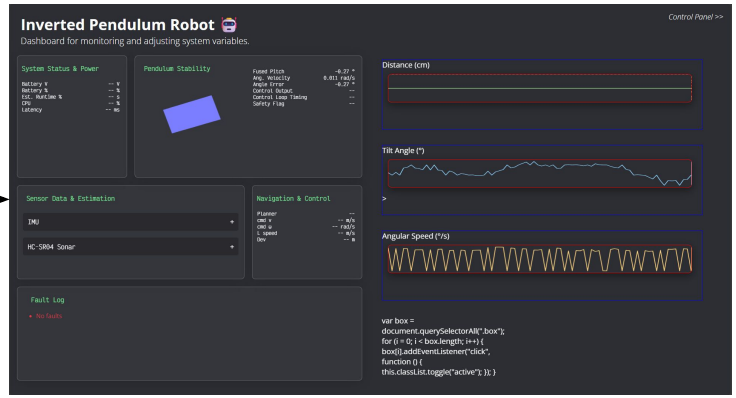
Isometric View

Integrated System Diagram



Integrated System Architecture

ESP-32 comm.



User Interface

Project Timeline

Subsystem Designs and Testing (complete by 2/9)	Integration of controls, sensors, and MCU (complete by 2/16)	Integration of MCU and motors (to complete by 2/20)	Full System Integration (complete by 3/6)	Full System Testing (complete by 3/20)	System Validation (complete by 4/1)	Demo and Final Report (complete by 4/20)

Controls Algorithm

Moustafa Khalil

Accomplishments since last update 50 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Integrated PID loop with motors • Integrated with sensors • Ongoing testing with IMU • Debugging Code	• Finish testing with IMU • Tune the LQR and PID gains • Make code more efficient

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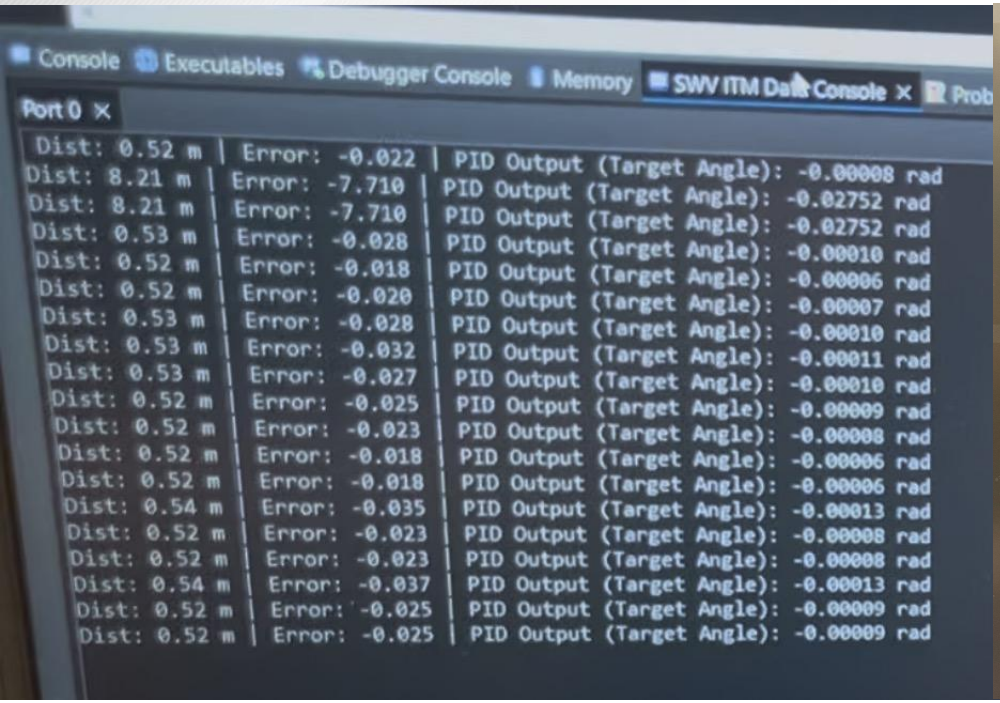
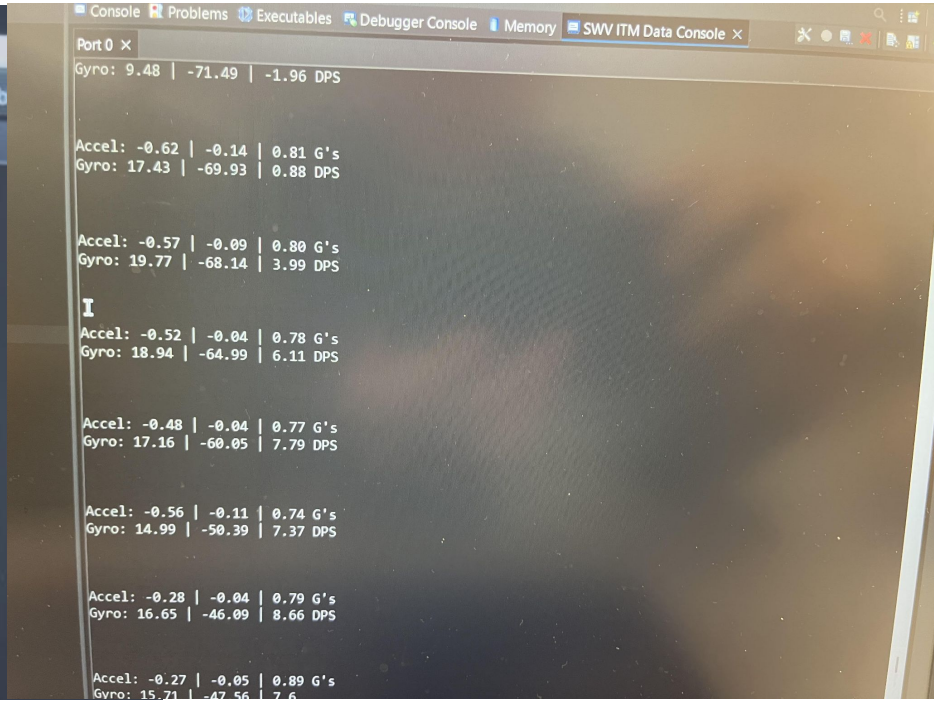
Controls Algorithm

Moustafa Khalil





Power & Mechanical

Kaitlynn Ly

Accomplishments since last update 19 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Completed mechanical and electrical integration of all subsystems • Ongoing IMU debugging integration	• Continue cross subsystem integration debugging and validation

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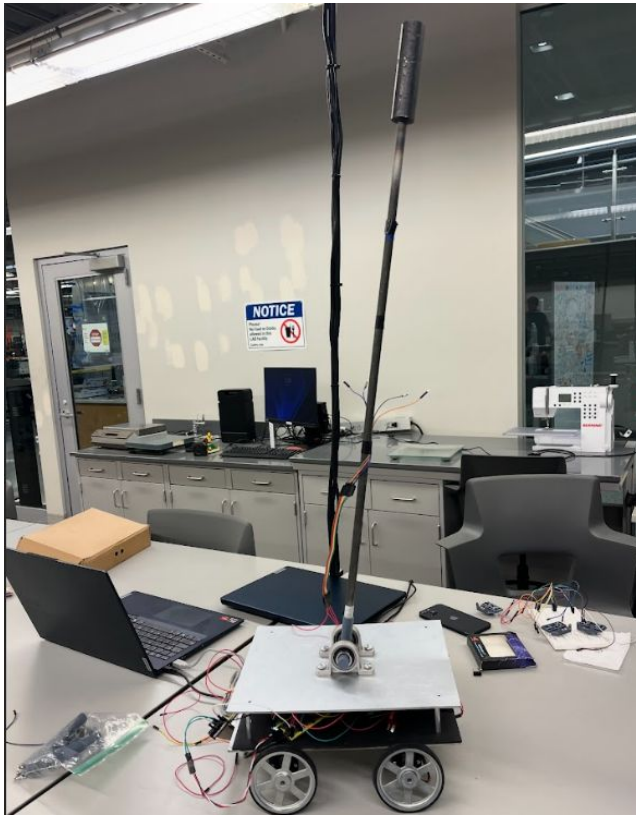
Power & Mechanical

Kaitlynn Ly



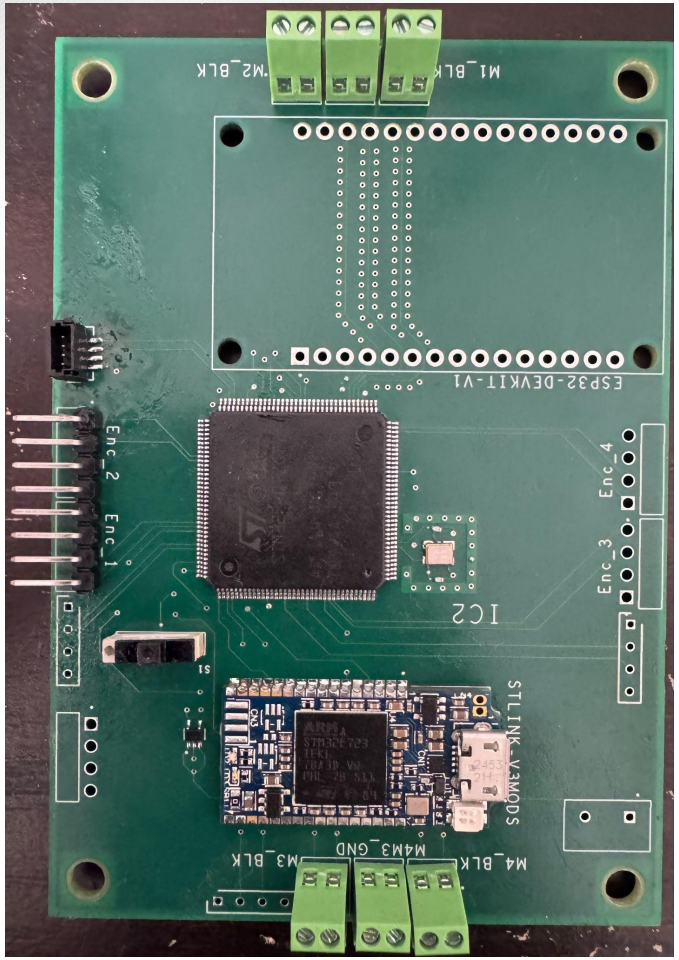
Logic MCU and PCB

Gage Fox

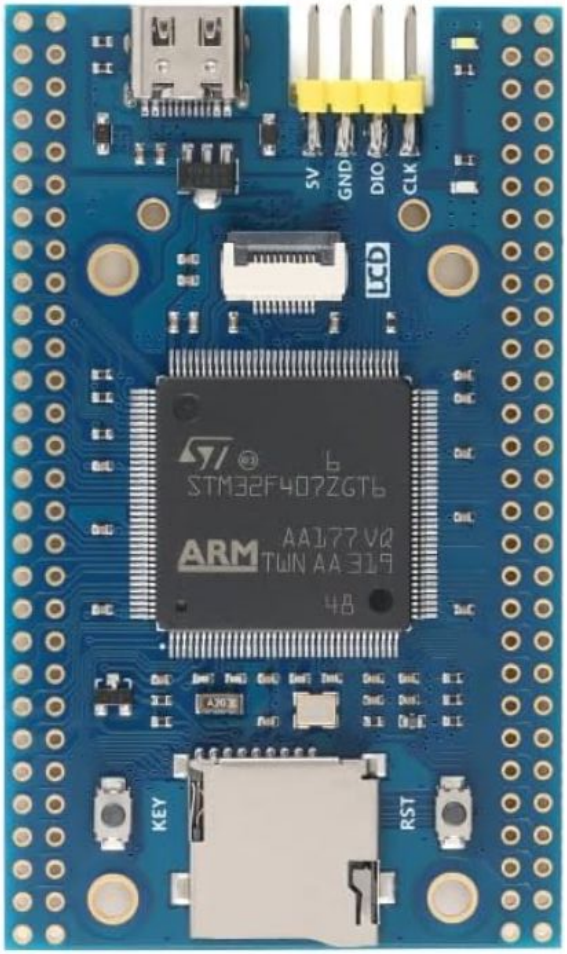
Accomplishments since last update 30 hrs of effort	Ongoing progress/problems and plans until the next presentation
• IMU Integration with Motors and Testing. • Code Modifications for Integration with Control Algorithm	• Soldering of Logic PCB for more reliable system • SPI Protocol with Wireless Module • DMA Implementation for Sensors and Motor Controls.

Logic MCU and PCB

Gage Fox



Custom PCB



Current DevBoard

Sensors & Navigation

Eric Chang

Accomplishments since last update 27 hrs of effort	Ongoing progress/problems and plans until the next presentation

● Git synchronization
● Debugging
  ● Timing conflicts
  ● Build configuration
● IMU stabilization
	● Merging code
	● Timing optimization

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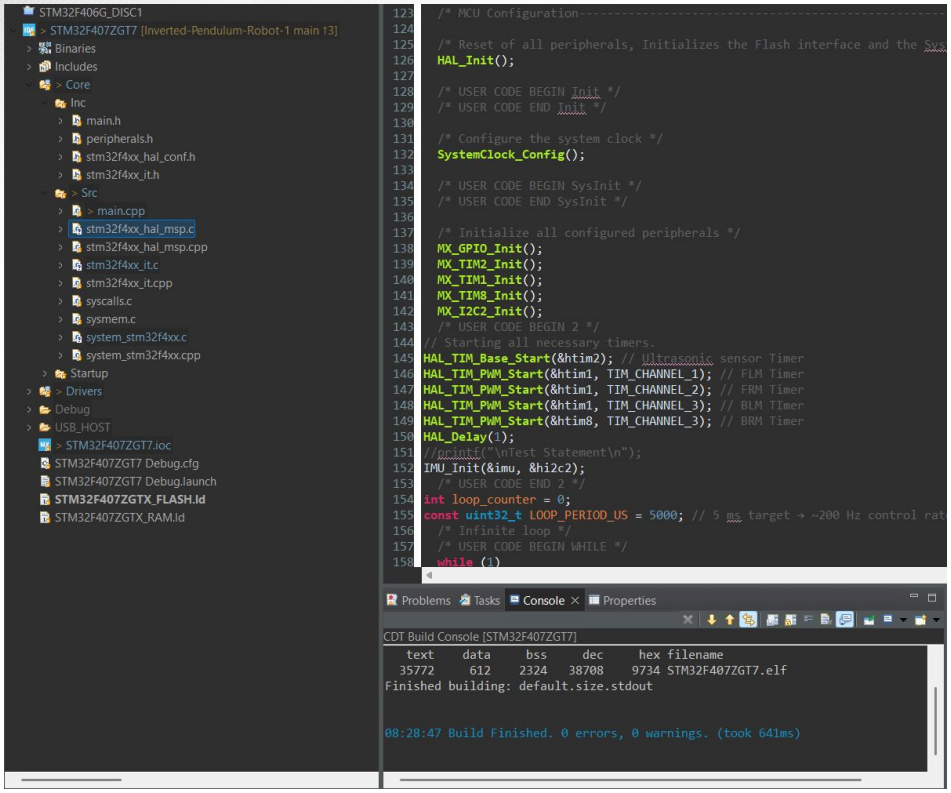
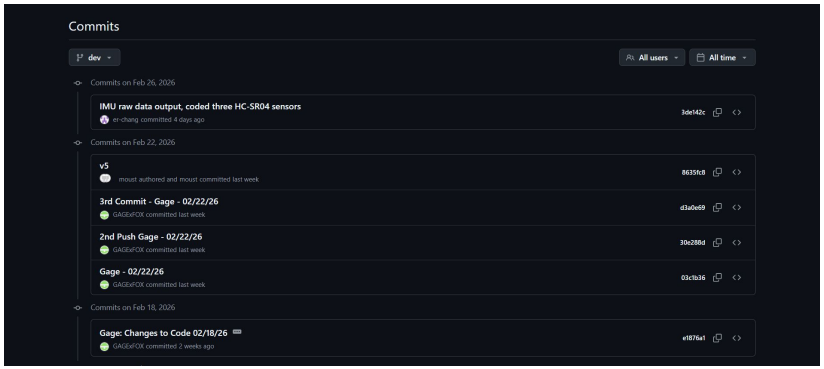
Sensors & Navigation

Eric Chang





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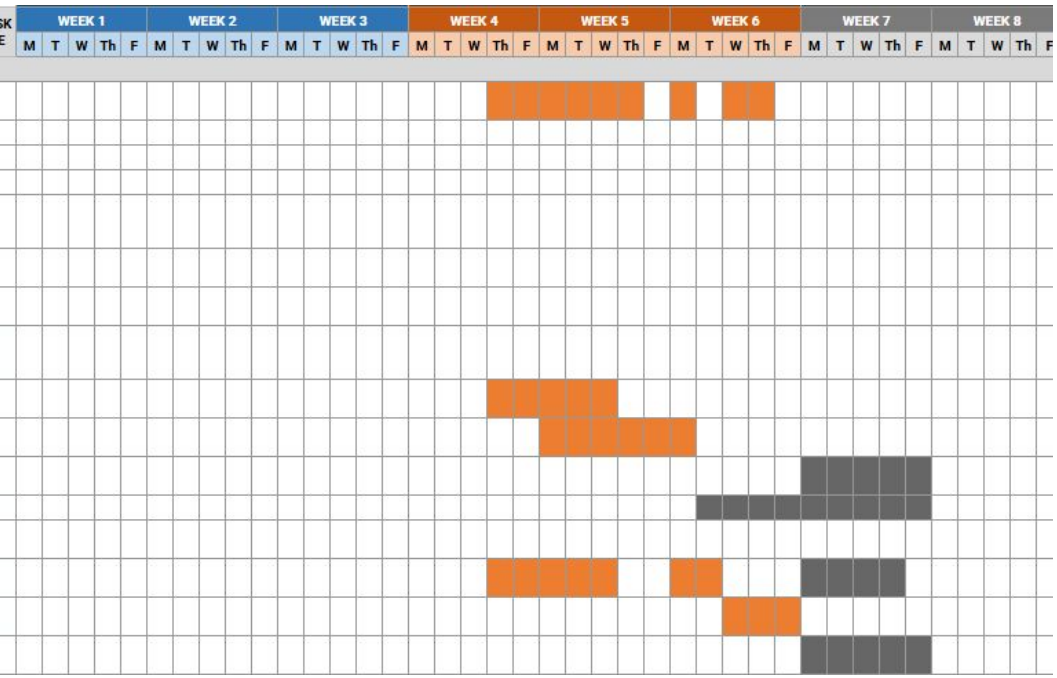
Execution Plan

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
2	Staged System Integration					
2.1	MCU and Controls Integration	Moustafa K.	9/15/25	9/19/25	4	50%
2.2	Sensor-to-MCU Physical Link	All	9/22/25	10/3/25	11	100%
2.3	Sensor Data received	All	9/24/25	9/24/25	0	100%
2.4	Power hooked on	Kaitlynn L.	1/13/26	1/20/26	7	100%
2.4.1	Verified consistent MCU startup when powered through the buck–LDO supply chain	Kaitlynn L.	2/9/25	2/26/26	4	100%
2.4.2	Validate motor driver oepration under full battery load	Kaitlynn L.	2/9/25	2/26/26	4	100%
2.4.3	Verify sensor noise introduced from switching regulator.	Kaitlynn L.	2/9/25	2/26/26	0	100%
2.4.4	Secure mounting of electrical components to reduce vibration induced wiring faults	Kaitlynn L.	2/9/25	3/7/25	0	75%
2.5	Test multiple sensors working simultaneously	Eric C.	2/5/26	2/12/26	7	100%
2.5.1	Measure and limit cross-talk/interference	Eric C.	2/5/26	2/12/26	7	100%
2.6	Integrate estimator output to control loop	Eric C.	2/12/26	2/19/26	7	100%
2.6.1	Test estimator timing between sensors	Eric C.	2/12/26	2/19/26	7	60%
2.7	Implement live fault notifications from subsystems in GUI	Eric C.	2/19/26	2/22/26	3	0%
2.8	Controls Algorithm cleanly reads sensor data	Moustafa K.	9/26/25	9/30/25	4	70%
2.9	Test Single-Component Interfacing with MCU	Gage F.	2/18/26	2/22/26	7	80%
2.9.1	Test simultaneous component interfacing with MCU	Gage F.	2/23/26	2/27/26	4	50%



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SYSTEM Validation Plan

Paragraph #	Task	Specifications and Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.	Logic Calculation Rate	Validate through timing tests on MCU execution cycle; must complete calculations in < 50ms	Measured using STLINKV2 Debugger	Tested	Gage
3.2.1.3	Data Transfer Rates	Measure message latency with time logs; must be < 100 ms difference	Measured using STLINKV2 Debugger and Analog Discovery Kit	Tested	Gage
3.2.3.1.7	Oscillator Precision	Validate clock stability through frequency tests; ensure deviation < 10ppm		Untested	Gage
3.2.3.1.4	Logic PCB Functionality	Verify that Logic PCB functions as intended without signal interferenece between components	Run test signals through all traces and read the resultant waveforms of all traces using an Analog Discovery Kit	Untested	Gage
3.2.3.1	I2C Communication	Verify that I2C Communication with IMU is functioning without error	Run the IMU alongside the MCU and compare the MCU's readings with the sensor data GUI.	Untested	Gage
3.2.5.1	SPI Master-Slave Communication	Verify proper Master-Slave communication between MCU and Wireless Module	Send and recieve test signals between the MCU (master) and Wireless Module (slave) and output these signals to their respective hosts.	Untested	Gage
3.2.3.1.7	Simultaneous Peripheral Controls and Comms	Validate that all peripherals operate as intended whilst being operated altogether.	Test run that the motors can be controlled with incoming data from all sensors, all at the same time using the MCU.	Untested	Gage
3.2.1.3	Sonar Read Distance	Validate using logged ultrasonic data (CSV) and IMU; confirm ≥95% obstacle detection at distances ≥2 cm and ±3 cm accuracy within 0.2-1.0 m	Use a meter stick to track distances, record N amount of trials at different positions, analyze collected results	Tested	Eric
3.2.1.3	Sensor Field-of-view Testing	Verify that the ultrasonic sensor coverage meets the ±15° design requirements	Use a meter stick to track distances, record N amount of trials at different positions, analyze collected results	Tested	Eric
3.2.3.1.5	IMU steady-state: bias, noise, drift	Verify IMU steady-state metrics: accel bias ±0.02 g, gyro bias ±0.5 °/s, accel noise σ ≤ 0.03–0.05 g, gyro noise σ ≤ 0.3–0.5 °/s, drift ≤1 °/min	Hold the IMU steady (without external disturbances) for an extended period, analyze collected results	Tested	Eric
3.2.5.1	Telemetry/GUI Latency	Mean latency < 200 ms with minimal packet loss	A clock will measure the time delay between the data sent from hardware and data received by the web interface	Tested	Eric
3.2.1.3	IMU and Sonar Dynamic Measurements	Mean error ≤2 ° during motion, mean distance error ≤ 3 cm	Keep the sensors in motion and compare recorded readings against reference values	Untested	Eric
3.2.4	Multi-sensor Coordination	Verify that sensor readings are ≥90% accurate while sensors are active, readings show minimal interference	Run all sensors simultaneously and check for false readings	Tested	Eric
3.2.5.1	Logging Test	Ensure the system is capable of logging all data during a complete run	Output sensor and control data to the web interface during motion and verify that data is accurate and timely	Untested	Eric
3.2.3.1.1	Motor Power Consumption	Measure motor current at multiple voltages under both free-spin and load conditions; confirm total motor power draw is ~40-45 W	Read motor current under different volatges	Tested	Kaitlynn
3.2.3.1	Verify Input Power Delivery	Measure system voltage under load; confirm range is within 11.1-12.6V	Use an oscilloscope to measure voltage under different loads	Tested	Kaitlynn
3.2.3.3	Buck Converter Regulation	Record output voltage under different loads; verify 5V stability	Verify 5V stability under multiple loads using an oscilloscope	Tested	Kaitlynn
3.2.3.4	LDO Validation	Measure regulated output with oscilloscope; confirm 3.3V stability	Measured 3.3V stability	Tested	Kaitlynn
3.2.3.1.2	MCU Power Delivery	Measure 3.3V at MCU during code execution; confirm stable operation	Measure 3.3V at MCU during code execution and confirm stable operation	Tested	Kaitlynn
3.2.3.1.6	Sensor Power Stablity	Measure sensor supply while streaming data; confirm no dropouts	Measure sensor supply while streaming data	Tested	Kaitlynn
3.2.3.5	Motor Driver Power	Measure driver input voltage/current during motor operation	Measure motor current at multiple voltages under both free-spin and load conditions; confirm total motor power draw is ~40-45 W	Tested	Kaitlynn
3.2.3.1	System Power Sequencing	Confirm safe startup; power logic rails first, then motor power	Confirm safe startup; power logic rails first, then motor power	Tested	Kaitlynn
3.2.3.1.7	Control Algorithm Validation	Validate stability across various operating conditions and confirm tuned gains achieve target response time	Verified that the controller was able to successfully balance the inverted pendulum	Tested	Moustafa
3.1.2.1.7	Initial Condition Response (LQR)	In simulation, initialize the pendulum at -10° offset (without PID input) and observe the angle v. time graph as it stabilizes	In simulation, initialize the pendulum at -10° offset (without PID input) and observe the angle v. time graph as it stabilizes	Tested	Moustafa
3.2.3.1.7	Step Response Tracking (PID)	Validate tracking performance by applying a 1.0 m step input; confirm rise time is < 2.5 s, overshoot is < 10%	Validate tracking performance by applying a 1.0 m step input; confirm rise time is < 2.5 s, overshoot is < 10%	Tested	Moustafa
3.2.3.1	Control Algorithm and MCU Integration	Able to run with the desired rise time, and give commands to rest of robot	Deploy compiled control loops to the MCU; verify PWM signal generation and command transmission frequency meet timing requirements for the desired rise time.	Tested	Moustafa
3.2.3.1.5	Control System and Sensor Integration	Able to cleanly read the data from sensors	Perform real-time polling of IMU and encoder registries; verify the control loop processes filtered data without lag or noise-induced jitter.	Untested	Moustafa
3.2.3.1.1	Controls System and Power Integration	Control subsystem successfully gives correct power to the robot to deal with various obstacles	Subject the robot to physical barriers; monitor current spikes and duty cycle adjustments to ensure the power system sustains high-torque demands.	Untested	Moustafa
3.2.1.1	Upright Pendulum Stability	Robot successfully balances pendulum in the upright position	Place robot upright, enable control lopop, and measure duration of stable balance	Untested	All
3.2.5.1	Over Tilt Failure Propagration	If tilt exceeds past 15 degrees . motors are immediately disable; system enters safe state and requires reset to enable	Intentionally tilt robot past 15 degrees and verify motor and safe state behavior	Untested	All
3.2.1.2	Maze Navigation Performance	Robot successfully navigates a maze from start to finish; maintains upright stability through navigation	Test navigation across multiple maze configurations to stimulate unknown demo conditions; record compelteion success and stablilty over repeated trials	Untested	All

Thank You!

Inverted Pendulum Robot 1

Any Questions?